

DESIGN DOCUMENTATION

Automotive Multi-ECU CAN Communication System with Crash Detection Module

1. Introduction

Modern vehicles use multiple Electronic Control Units (ECUs) to control and monitor various automotive functions such as engine management, braking systems, infotainment, transmission control, battery monitoring, and safety systems. These ECUs communicate with each other using the Controller Area Network (CAN) protocol.

This project implements an Automotive Multi-ECU CAN Communication System with a Crash Detection Module using the STM32F407G Discovery Board. The crash detection module uses a KY-002 shock sensor to detect sudden vibration or impact conditions. When a crash event is detected, the STM32 microcontroller activates a warning LED and buzzer while transmitting a CAN message over the CAN bus using the TJA1050 CAN transceiver module.

The CAN communication is monitored using CANeasy software to simulate communication between multiple automotive ECUs.

2. Project Objective

The objectives of this project are:

- To simulate an automotive multi-ECU CAN communication network
 - To detect crash events using the KY-031 shock sensor
 - To process crash events using EXTI interrupt mechanism
 - To activate LED and buzzer alerts during crash detection
 - To transmit crash status through CAN communication
 - To monitor CAN messages using CANeasy software
 - To demonstrate embedded automotive communication architecture
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3. System Overview

The project simulates a vehicle CAN network consisting of multiple ECUs.

ECU Name	Function
Engine ECU	Engine parameter monitoring
ABS ECU	Brake system monitoring
Body Control Module (BCM)	Vehicle body electronics
Infotainment ECU	Display and multimedia system
Transmission ECU	Gear and transmission monitoring
Battery ECU	Battery monitoring system
Crash Detection ECU	Crash sensing and CAN transmission

The Crash Detection ECU is implemented using the STM32F407G-DISC1 development board.

4. Hardware Components

Sl No	Component	Purpose
1	STM32F407G-DISC1 Discovery Board	Main microcontroller
2	KY-031 Knock Sensor	Crash and vibration detection
3	TJA1050 CAN Transceiver Module	CAN bus communication
4	LED	Crash indication
5	Buzzer	Audio alert
6	Jumper Wires	Hardware connections
7	Breadboard	Circuit prototyping
8	USB Cable	Power supply and programming

5. Software Requirements

Software	Purpose
STM32CubeIDE	Firmware development
CANeasy	CAN message monitoring
Embedded C	Firmware programming
STM32 HAL Library	Peripheral configuration

6. Pin Configuration

Component	STM32 Pin	Description
KY-031 Knock Sensor	PA1	EXTI interrupt input
LED	PD14	Crash indication output
Buzzer	PA5	Audio alert output
CAN TX	PD1	CAN transmission
CAN RX	PD0	CAN reception

7. Working Principle

During normal operation, the KY-002 shock sensor continuously monitors vibration conditions. The sensor output is connected to PA0 configured as an EXTI interrupt input.

When vibration or sudden impact occurs:

KY-031 sensor detects vibration.

EXTI interrupt is triggered on PA1.

STM32 processes the crash event.

LED connected to PD12 starts blinking continuously.

Buzzer connected to PD13 activates for a few seconds.

CAN frame is transmitted using CAN ID 0x100.

CAN data byte 0x01 is sent indicating crash detection.

CANeasy monitors and displays the CAN message.

During normal condition:

CAN data byte 0x00 indicates normal status.

8. CAN Communication Design

8.1 CAN Configuration

Parameter	Value
CAN Protocol	CAN 2.0
CAN Transceiver	TJA1050
CAN ID	0x100
Data Length	1 Byte
Communication Type	Broadcast
CAN Speed	500 kbps

8.2 CAN Data Format

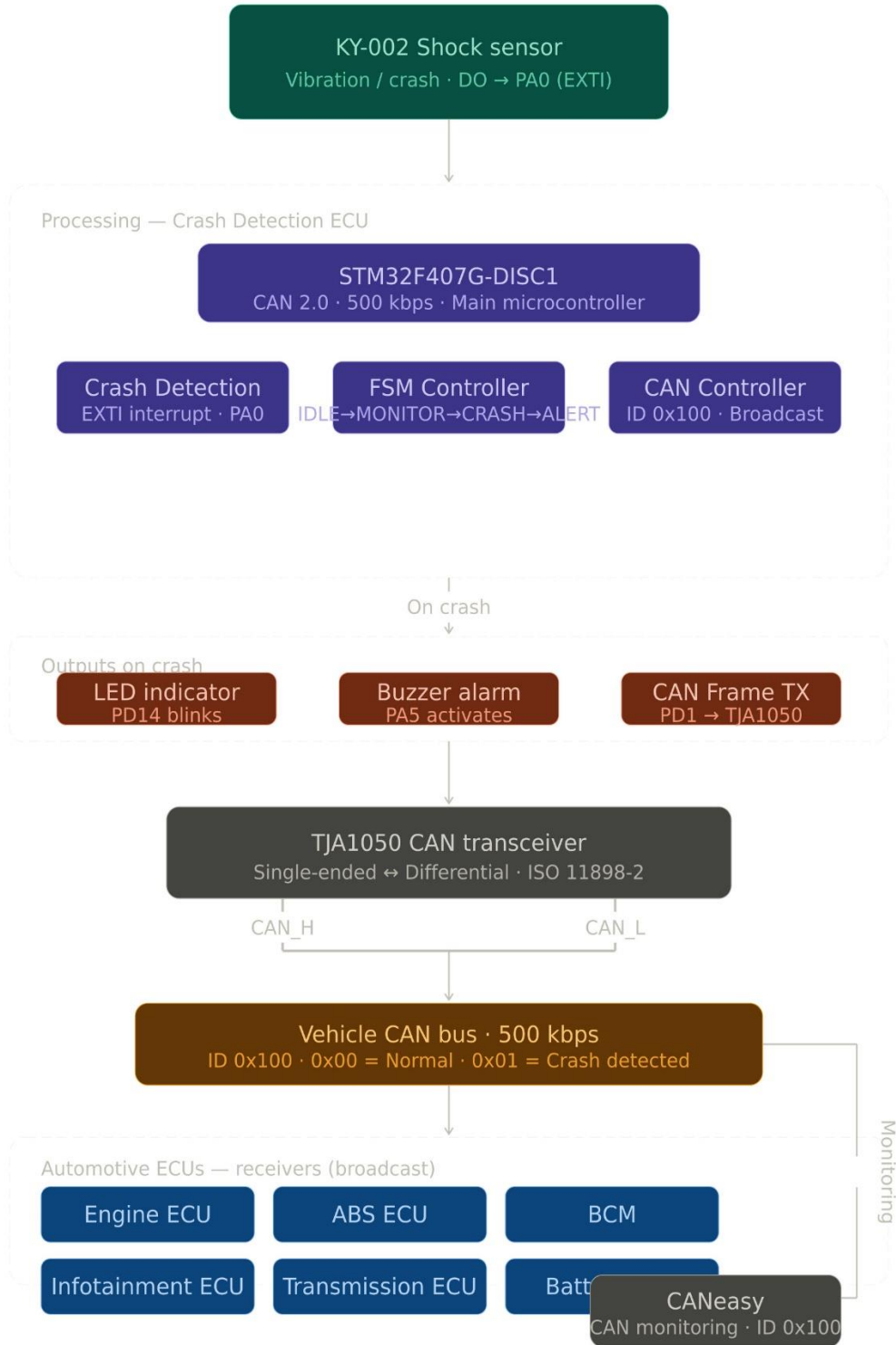
CAN ID	Data Byte	Meaning
0x100	0x00	Normal Condition
0x100	0x01	Crash Detected

9. Finite State Machine (FSM)

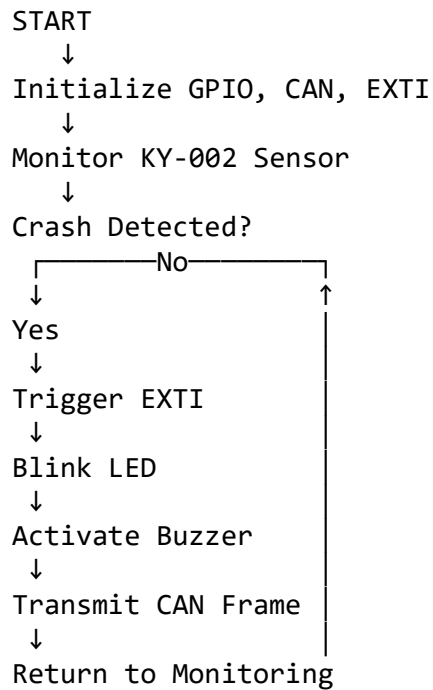
The Crash Detection ECU operates using the following states:

State	Description
IDLE	System initialization state
MONITORING	Monitoring sensor input
CRASH_DETECTED	Crash identified through EXTI interrupt
ALERT_TRANSMISSION	LED blinking, buzzer activation, and CAN transmission

10. Block Diagram



11. Flowchart



12. Software Architecture

File Name	Function
main.c	Main application
gpio.c	GPIO configuration
can.c	CAN communication
exti.c	External interrupt handling
crash_handler.c	Crash detection logic

13. Testing and Validation

Test ID	Test Description	Expected Result
T1	Sensor vibration test	EXTI interrupt triggered
T2	LED test	LED blinks continuously
T3	Buzzer test	Buzzer activates
T4	CAN transmission test	CAN ID 0x100 transmitted
T5	CAN data validation	0x01 transmitted during crash
T6	CANeasy monitoring	Message visible in CANeasy

14. Advantages

- Real-time crash detection
- Fast CAN communication
- Interrupt-based event handling
- Simple embedded implementation
- Automotive communication simulation
- Low-cost hardware implementation
- Internship-level automotive project

15. Future Scope

- Real airbag deployment module integration
- Multiple crash sensor support
- GPS emergency notification
- GSM-based accident alert system
- Integration with additional ECUs
- Advanced automotive diagnostics
- CAN FD implementation

16. Conclusion

This project successfully demonstrates an Automotive Multi-ECU CAN Communication System with Crash Detection Module using the STM32F407G Discovery Board. The system detects vibration using the KY-031 Knock sensor and processes crash events using EXTI interrupt handling. During crash detection, LED blinking and buzzer activation are performed while a CAN message is transmitted through the TJA1050 CAN transceiver.

The project demonstrates the fundamentals of automotive embedded systems, CAN communication, interrupt handling, and ECU interaction, making it suitable for internship-level automotive embedded applications.